

CO2 ASSESSMENT TOOL 1 LEVEL 1

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
C PROGRAMMING

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CO2 - AT1 - LEVEL-1

Level Exam 02 Sep 2026, 02:30 PM

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Result Available
Your score: 91 / 100

QUESTIONS

LEVEL 1 — EASY

Q1
Number

Q2
Number

Q3
Number

Q4
Pattern

4 / 4 saved

Level 1 — Easy

Palindrome Pattern (E) — Pattern

Submitted

Problem: Print the following Palindrome Pattern

Sample Input:
(None)Sample Output:
1
121
12321
1234321
123454321Test Cases:
===TEST CASE 1===
Input:
(None)
Expected Output:
1
121
12321
1234321
123454321

===TEST CASE 2===
Input:
(None)
Expected Output:
1
121
12321
1234321
123454321

Your Code

```
1 #include<stdio.h>;
2 int main() {
3     int n=5;
4     for(int i=1;i<=n;i++){
5         for(int j=1;j<=i;j++){
6             printf("%d",j);
7         }
8         for(int j=i-1;j>=1;j--){
9             printf("%d",j);
10        }
11        printf("\n");
12    }
13    return 0;
14 }
```

Output

```
1
121
12321
1234321
123454321
```

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Result Available
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QUESTIONS

LEVEL 1 — EASY

- Q1
Number
- Q2
Number
- Q3
Number
- Q4
Pattern

4 / 4 saved

Level 1 — Easy digit (E) — Number

Submitted

Problem: Write a program to count the number of digits in a number entered by the user.

Sample Input:
Enter a number: 12345Sample Output:
Number of digits: 5Test Cases:
===TEST CASE 1===
Input:
Enter a number: 12345
Expected Output:
Number of digits: 5===TEST CASE 2===
Input:
9876
Expected Output:
Number of digits: 4===TEST CASE 3===
Input:
0
Expected Output:
Number of digits: 1

Your Code

```
1 #include<stdio.h>
2 int main(){
3     long long n;
4     int count=0;
5     printf("Enter a number:");
6     scanf("%lld",&n);
7
8     if(n==0){
9         count=1;
10    }else{
11        if(n<0)n=-n;
12        while(n!=0){
13            n/=10;
14            count++;
15        }
16    }
17    printf("Number of digits:%d\n",count);
18    return 0;
19 }
```

Output

Enter a number: Number of digits:5

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QUESTIONS

LEVEL 1 — EASY

- Q1
Number
- Q2
Number
- Q3
Number
- Q4
Pattern

4 / 4 saved

Level 1 — Easy sum (E) — Number

✓ Submitted

Problem: Write a program to repeatedly sum the digits of a number until the result is a single digit.

Sample Input:
Enter a number: 9875Sample Output:
Single digit sum is 2Test Cases:
===TEST CASE 1===
Input:
Enter a number: 9875
Expected Output:
Single digit sum is 2===TEST CASE 2===
Input:
Enter a number: 9875
Expected Output:
Single digit sum is 2

Your Code

```
1 #include<stdio.h>
2 int main(){
3     long long n;
4     int sum;
5     printf("Enter a number: ");
6     scanf("%lld",&n);
7     if(n<0)n=-n;
8
9     while(n>=10){
10        sum=0;
11        while(n>0){
12            sum+=n%10;
13            n/=10;
14        }
15        n=sum;
16    }
17    printf("Single digit sum is %lld",n);
18    return 0;
19 }
```

Output

Enter a number: Single digit sum is 2

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Result Available
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QUESTIONS

LEVEL 1 — EASY

- Q1
Number
- Q2
Number
- Q3
Number
- Q4
Pattern

4 / 4 saved

Level 1 — Easy Palindrome Pattern (E) — Pattern

Submitted

Problem: Print the following Palindrome Pattern

Sample Input:
(None)Sample Output:
1
121
12321
1234321
123454321Test Cases:
---TEST CASE 1---
Input:
(None)
Expected Output:
1
121
12321
1234321
123454321

---TEST CASE 2---
Input:
(None)
Expected Output:
1
121
12321
1234321
123454321

Your Code

```
1 #include<stdio.h>
2 int main() {
3     int n=5;
4     for(int i=1;i<=n;i++){
5         for(int j=1;j<=i;j++){
6             printf("%d",j);
7         }
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10        }
11        printf("\n");
12    }
13    return 0;
14 }
```

Output

```
1
121
12321
1234321
123454321
```

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